

RUKAIYA KHAN

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SKILLS

- **Languages:** Java • Python • JavaScript (ES6+) • TypeScript
- **Frontend Frameworks & UI:** Next.js • React.js • React Hooks • Redux • HTML • CSS • Tailwind • SCSS • SASS
- **Databases:** MySQL • PostgreSQL • Redis • MongoDB • Firebase • Amazon RDS • Vector Databases (FAISS, Pinecone, Chroma)
- **LLM & AI Engineering:** Retrieval-Augmented Generation (RAG) • LangChain • LangGraph • Model Context Protocol (MCP) • Vector Search • Prompt Engineering • NER • Sequence Labeling • Text Classification • Agentic Systems & Workflows • Tool Calling • Memory & State Management • LLM-backed Serverless Architectures
- **Backend & APIs:** Node.js • Express.js • Nest.js • REST APIs • GraphQL • WebSockets • CORS Handling • Fast API • Flask
- **Cloud & DevOps:** Git • Docker • Linux • AWS Serverless (Lambda, API Gateway, API Kinesis, SQS, Step Functions, S3, EventBridge, CodePipeline, CodeBuild, CloudWatch, Cognito) • CI/CD • GitHub Actions • Version Control (Git/GitHub)
- **Other Tools & Technologies:** Claude Code • Microservices • Puppeteer • Chromium • AG Grid • Bootstrap • Material UI • Agile Methodologies • Kanban • Jupyter Notebooks • JSON/XML Parsing • Navigator AI

EDUCATION

Masters in Computer Science

University of Florida

Gainesville, August 2025 - May 2027

Bachelors in Computer Science

C.V. Raman Global University

Bhubaneswar, August 2019 - May 2023

CGPA - 9.0/10.0

WORK EXPERIENCE

SOFTWARE DEVELOPER

Ncompass

Chennai, January 2023 - August 2025

- Architected multi-tenancy support for a global e-commerce platform handling 30+ tenants and 20k+ customers per tenant, each with isolated databases and unique feature configurations (B2B workflows, pre-orders, auctions).
- Led Node.js migrations v14 → v16 → v20 across the SaaS application, ensuring zero downtime, full compatibility, and leveraging latest performance and security improvements.
- Automated client onboarding workflows using AWS Step Functions, reducing processing times by 80% and supporting onboarding for thousands of customers per tenant.
- Cached frequent configurations in Redis, cutting database load and improving API performance and response times by 90%.

SOFTWARE DEVELOPER INTERN

New Delhi, May 2020 - December 2020

Xaltam Technologies

- Developed and maintained front-end websites and web applications, ensuring responsive design and user-friendly interfaces.

PROJECTS

CAREER OUTREACH AUTOMATION ENGINE

- Automated personalized job outreach by scraping JDs, using LiteLLM to select the best-fit resume and draft tailored cold emails, then syncing drafts to Notion for approval before sending via Gmail SMTP with resume attachments.
- Implemented vector embeddings for few-shot example retrieval and automated multi-stage follow-ups, with Notion serving as the sole UI for managing the entire pipeline — job queue, drafts, approvals, and send status.

AI-POWERED EMAIL THREAT DETECTION & LLM MONITORING

- Built a full-stack email security platform with FastAPI and React, integrating Google OAuth and Gmail API for real-time inbox threat analysis using UFL's LiteLLM proxy to classify phishing, spam, and legitimate emails with confidence scoring.
- Engineered a metrics dashboard with live latency tracking, token usage, and cost estimation, backed by PostgreSQL and a caching layer for duplicate detection across multi-user workflows.

OPINION INTELLIGENCE SYSTEM

- Built full-stack Chrome extension with FastAPI backend to analyze Reddit discussions, integrating LLM-powered AI pipeline for real-time opinion extraction and sentiment analysis.
- Developed multi-step clustering and NLP pipeline extracting consensus viewpoints, debate topics, sentiment classification, and missing perspectives from thousands of comment threads.

FINANCE HANDLER AI BASED DASHBOARD

- Built a responsive finance dashboard with animated stat cards, count-up numbers, staggered Framer Motion reveals, interactive Recharts visualizations (area, donut, bar charts), and a dark/light theme system using CSS variables
- Integrated a context-aware AI chat panel via GPT-3.5 LiteLLM proxy with live spending insights, real-time transaction state using React Context, color-coded budget alerts, and <350ms page transitions with mobile-adaptive navigation

AWARDS AND ACTIVITIES

Google Developer Student Clubs Lead, CVRGU, GDSC

July 2021 - July 2022

- Volunteered with GDSC-CVRGU, growing the community to 800+ members, organizing 8+ events, and mentoring peers

Smart India Hackathon Finalist, Ministry of Education's Innovation Cell

February 2022

- Published patent (Application ID: 202531060664) for an AI-powered, interactive virtual campus exploration system integrating geotagging on Bharat Map with minimal-equipment VR tours of educational institutions

Automated Crop Disease Detection and Management System

January 2022 - May 2023

- Published patent (Application ID:202531061085) for an AI-powered solution that detects crop diseases and provides management recommendations to support precision agriculture